CEHv13 Module 20

Cryptography Lab Report

Certified Ethical Hacker v13 | 8 November 2026 | Result: Passed (3/5), duration 1 hour 32 minutes

1. Executive summary

The Module 20 lab was passed with a score of 3/5 in 1 hour 32 minutes. Three hands-on tasks covered the core cryptography skills a Certified Ethical Hacker must recognise: hashing, encryption and secure web transport.

- **Hashing:** a multi-layer recipe (MD5, SHA1, HMAC) was built in CyberChef on a sample account number, and the Set Breakpoint control was identified as the way to pause a recipe before it runs.
- **Encryption:** a confidential text file was protected with CryptoForge, which writes encrypted files with the `.cfe` extension.
- **PKI and HTTPS:** a self-signed certificate was created in IIS and bound to www.goodshopping.com on 10.10.1.19, with HTTPS served on port 443.

2. Lab environment and tools

The tasks ran across a Windows 11 workstation and a Windows Server 2019 host named SERVER2019, using tools supplied in the CEH-Tools share.

Tool	Host	Purpose
CyberChef (browser)	Windows 11	Chain hash and HMAC operations on sample text
CryptoForge	Windows 11	Encrypt and decrypt a confidential file and text
IIS Manager	SERVER2019	Create a self-signed certificate and add an HTTPS binding
Notepad	Windows 11	View the plaintext files before and after encryption

3. Task 1: Multi-layer hashing with CyberChef

A three-layer recipe turned the 31-character string "My Account number is 0234569198" into a single 32-character hexadecimal value, `bcabf7b4fd5f95e706d933e8119dab01`. Each operation takes the output of the one above it as its input.

Layer	Operation	Setting
1	MD5	Default
2	SHA1	Rounds: 80
3	HMAC	Key: 12 (HEX); hashing function: MD5

Every step in the recipe carries a disable button and a pause button. The pause button is **Set Breakpoint**, which stops execution before that step runs so the intermediate value can be inspected. This was the answer recorded for the lab question.

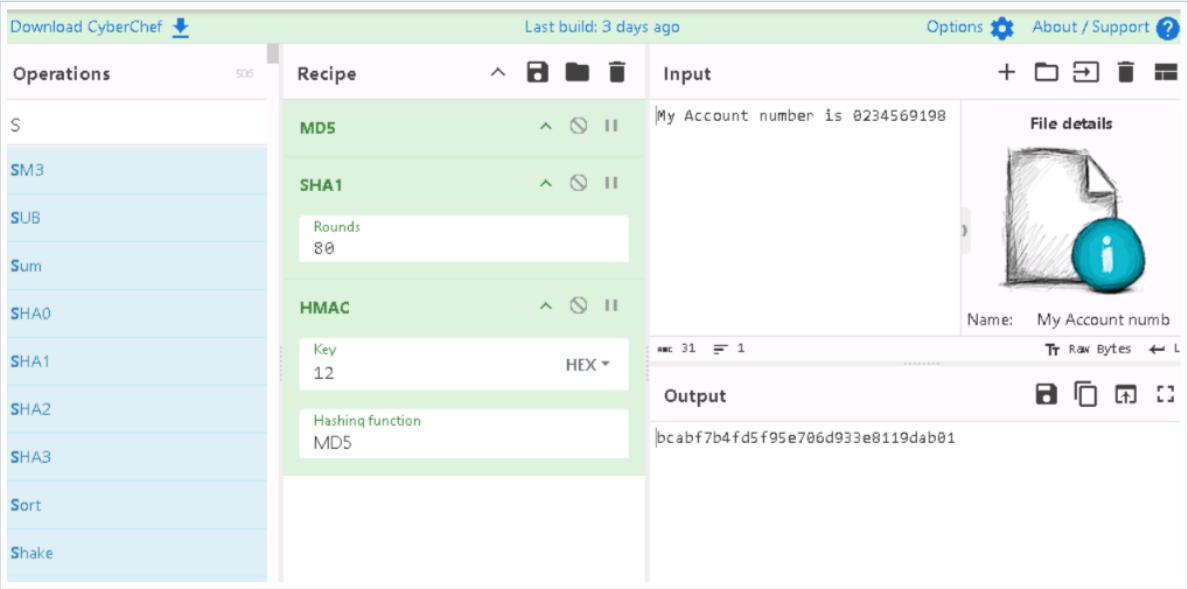


Figure 1. CyberChef recipe (MD5, SHA1, HMAC-MD5) applied to the input text, with the final output.

Two points matter for the exam and for practice. A hash is one-way, so the original account number cannot be recovered from the output. An HMAC adds a secret key, so the result depends on both the message and the key, which protects integrity and authenticity as well.

4. Task 2: File encryption with CryptoForge

CryptoForge protected the sensitive contents of `Confidential.txt`, and files it encrypts take the `.cfe` extension. This was the answer recorded for the lab question.

The source file is 115 bytes, dated 12/10/2019, and sits beside `CryptoForge.exe` (8,005 KB) in the CEH-Tools share. It holds three lines of masked financial data: a Visa card number, a bank account number and a card PIN.

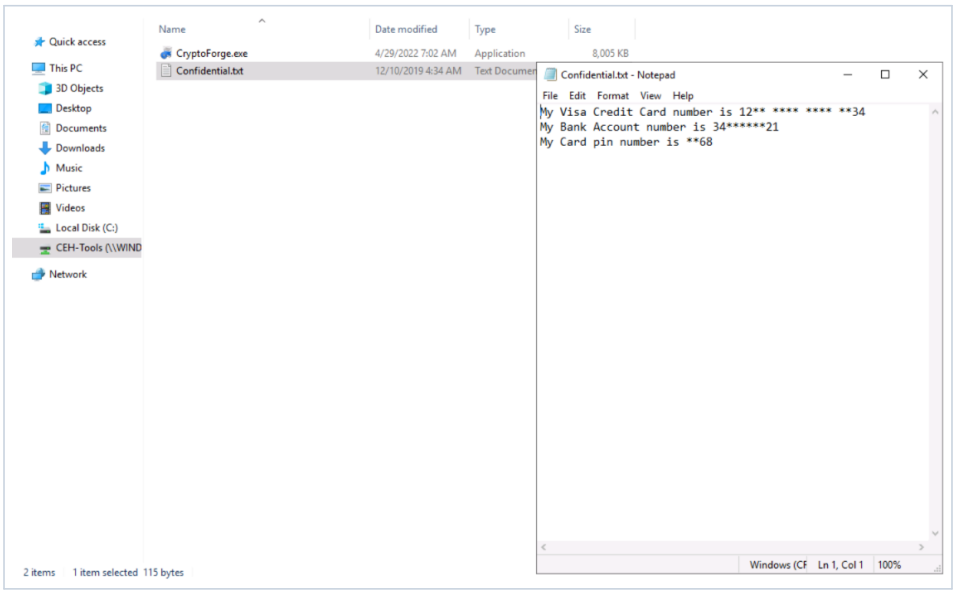


Figure 2. The source file Confidential.txt (masked values) and the CryptoForge installer.

1. Open `Confidential.txt` in Notepad and confirm the plaintext.
2. Start CryptoForge and set a passphrase with the Passphrase button.

3. Select Encrypt. The text becomes an armoured block that starts with a CRYPTOFORGE BEGIN BLOCK header and ends with a matching END BLOCK line, so the content reads as unreadable ciphertext.
4. Select Decrypt with the same passphrase. The original line, "My Visa Credit Card number is 12** **** **** **34", returns intact.

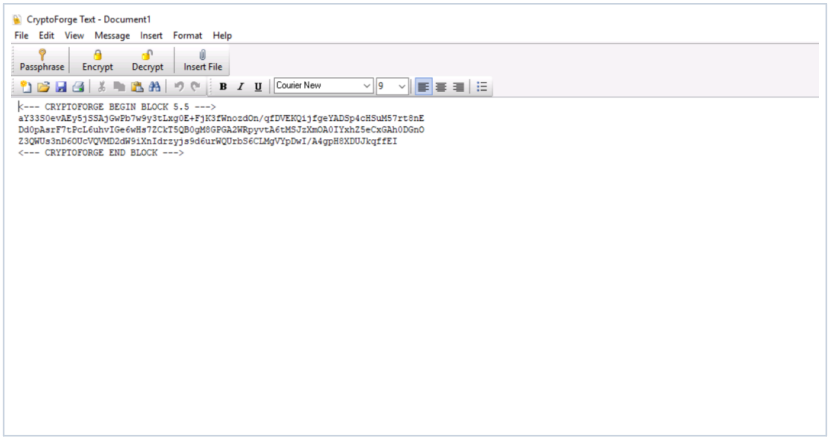


Figure 3. The encrypted block produced by CryptoForge: unreadable ciphertext between BEGIN and END markers.

Figure 4. After decryption the original line is restored.

This shows the defining property of symmetric encryption: the same secret passphrase protects and recovers the data, so protecting that passphrase is the whole security of the file.

5. Task 3: Self-signed certificate and HTTPS in IIS

HTTPS for www.goodshopping.com now runs on port 443, secured by a self-signed certificate named GoodShopping. This was the answer recorded for the lab question.

In IIS Manager on SERVER2019, the Server Certificates page lists the new certificate. It is issued to and by Server2019 and expires on 11/6/2027. A self-signed certificate is issued by the server to itself, so no outside authority vouches for it.

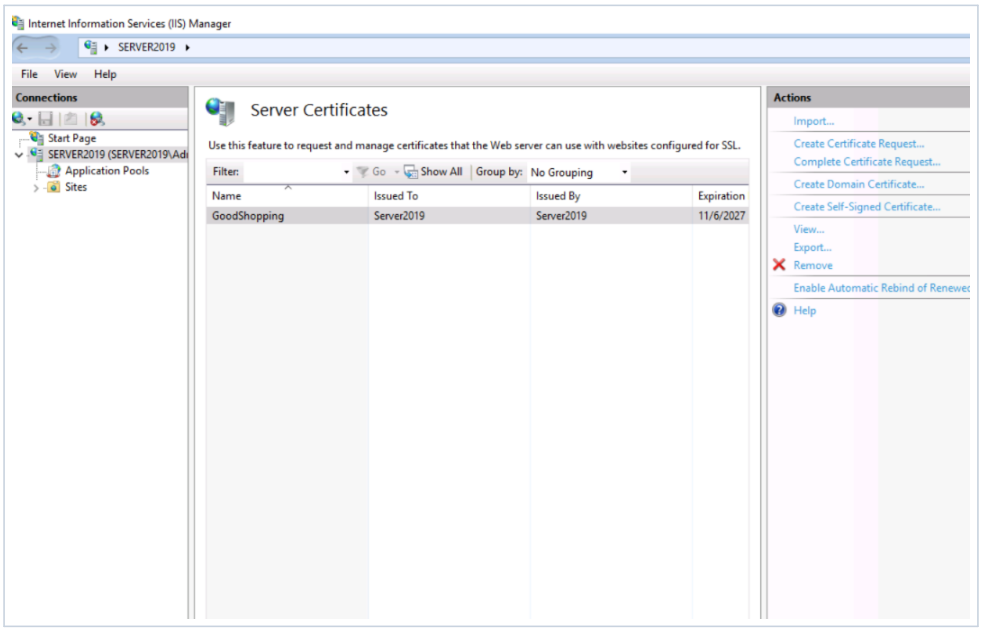


Figure 5. The GoodShopping self-signed certificate in IIS Server Certificates.

The GoodShopping site already had an HTTP binding for its host name on port 80 at 10.10.1.19. A second binding added the secure listener:

Setting	Value
Type	https
IP address	10.10.1.19
Port	443
Host name	www.goodshopping.com
Require Server Name Indication	Unchecked
Disable HTTP/2, Disable OCSP Stapling	Unchecked
SSL certificate	GoodShopping

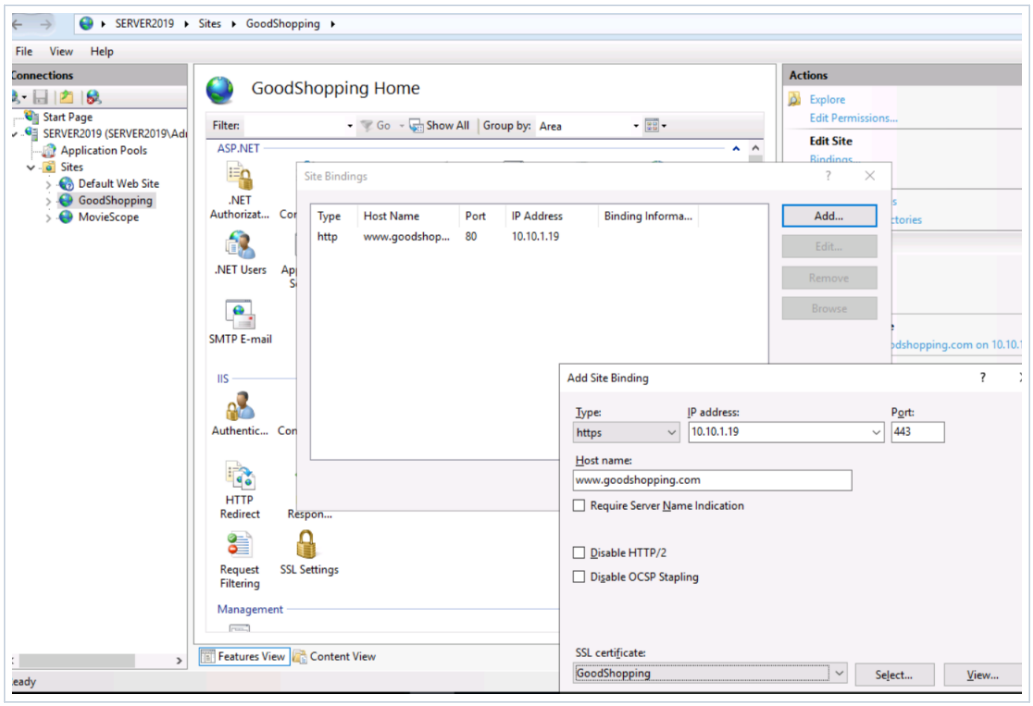


Figure 6. The https binding for www.goodshopping.com on 10.10.1.19, port 443.

The server also hosts the sites Default Web Site and MovieScope, which are unaffected by this binding.

6. Results

The assessment shows a pass at 3/5 after 1 hour 32 minutes, and each of the three questions listed was marked correct for 1 point.

Task	Question	Answer recorded	Score
1	Which recipe button pauses execution before the step runs?	Set Breakpoint	1
2	What is the extension of the file CryptoForge encrypts?	.cfe	1
3	On which port does HTTPS run for www.goodshopping.com?	443	1

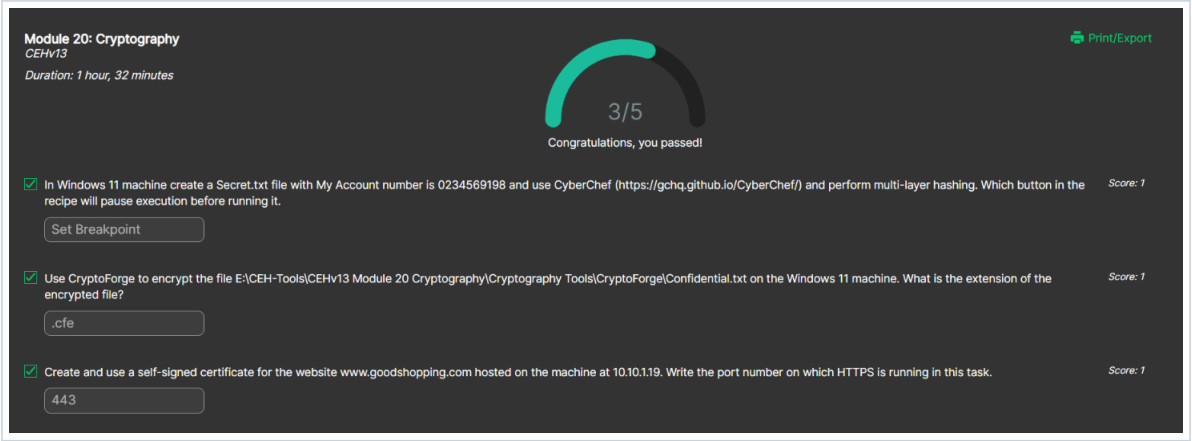


Figure 7. Module 20 result: passed, 3/5, duration 1 hour 32 minutes.

7. Security observations and recommendations

The lab settings suit a training environment. Four changes would be needed before they protect real data.

- **Replace legacy hashes.** MD5 and SHA1 no longer resist collision attacks. Use SHA-256 or SHA-3 for integrity checks, and use a long random HMAC key instead of the short demonstration key.
- **Protect the passphrase and the plaintext.** CryptoForge is only as strong as its passphrase. Choose a long one, store it separately from the data, and securely delete the original `Confidential.txt` once it is encrypted.
- **Use a trusted certificate in production.** Browsers warn on self-signed certificates because no authority vouches for them. Obtain a certificate from a trusted CA for public sites, and track the 11/6/2027 expiry for renewal.
- **Force HTTPS.** The port 80 binding still serves the site in plain text. Redirect HTTP to HTTPS, and restrict the server to current TLS versions.

8. Conclusion

The lab covered the three layers of practical cryptography: integrity through hashing and HMAC, confidentiality through symmetric encryption, and trust and transport security through certificates and HTTPS. Each tool produced the expected result, and all recorded answers were correct. The main lesson is that the algorithm is only part of the defence: key and passphrase handling, certificate trust and correct server configuration decide whether the protection holds.